

## GUÍA 7

7.1

$\{N(t), t \geq 0\} \rightarrow$  proceso de conteo asociado a un proceso de Poisson de intensidad  $\lambda = 2 = \mu$

$N(a,b) = N(b) - N(a)$  el incremento del proceso en el intervalo  $(a,b)$

a)  $P(N(1)=0)$

$t=1 \quad \lambda=2$   
 $\mu=2$

Nº de eventos

PP  $\rightarrow \lambda = 2$

$N(t) \sim Po(2t)$

$P(N(1)=0) = \frac{2^0 e^{-2}}{0!}$

$\mu=2$   
 $m=0$

$P(N=m) = P(N=m) = \frac{\mu^m e^{-\mu}}{m!}$

$P(N(1)=0) = e^{-2}$

b)  $P(N(1,2)=1) =$

$N(1,2) = N(2) - N(1)$

$m=1 \quad \mu = 2 \cdot 1 = 2 \cdot 1$   
 $\downarrow$   
long intervalo

$P(N(1,2)=1) = \frac{2^1 e^{-2}}{1!} = 2e^{-2}$

c)  $P(N(1)=0, N(1,2)=1, N(2,4)=2) = P(N(1)=0) \cdot P(N(1,2)=1) \cdot P(N(2,4)=2)$

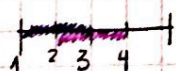
$\downarrow$   
como son ind  
por contar arriba  
en intervalos disjuntos

$= e^{-2} \cdot 2e^{-2} \cdot \frac{4^2 e^{-4}}{2!} = 2e^{-4} \cdot 8e^{-4} = 16e^{-8}$

dato:  $e^{-4} \cdot e^{-4} = e^{-8}$

DISTRIBUCION INTERVALOS

d)  $\text{COV}(N(1,3), N(2,4)) \stackrel{\downarrow}{=} \text{COV}[(N(1,2) + N(2,3)), (N(2,3) + N(3,4))]$



$\uparrow$   
hago distribucion  
 $= \text{COV}(N(1,2), N(2,3)) + \text{COV}(N(1,2), N(3,4)) +$

$\text{COV}(N(2,3), N(2,3)) + \text{COV}(N(2,3), N(3,4))$

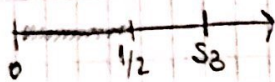
$= \text{COV}(N(2,3), N(2,3)) = \text{VAR}(N(2,3)) = 2 \cdot 1 = 2$

$\text{VAR}(N) = \mu$

4) si el evento o punto del proceso

e) Sea  $S_3$  el tiempo de espera hasta q' ocurre el 3er evento.

$$P(S_3 > 1/2) = P(N(1/2) < 3)$$



$S_3 > 1/2 \rightarrow$  esperar + de  $1/2$  hasta 3er evento.

$$= P(N(1/2) \leq 2)$$

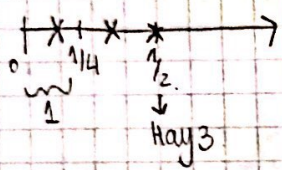
estatismo porque el 3mo entra

porque se que en ese intervalo, 2 llegaron seguro.

$$= \sum_{i=0}^2 \frac{(-2 \cdot \frac{1}{2})^i}{i!} e^{-1} =$$

$$= e^{-1} + e^{-1} + \frac{1}{2} e^{-1} = \frac{5}{2} e^{-1}$$

a)  $P(N(\frac{1}{4})=1 | S_3=\frac{1}{2}) = P(N(\frac{1}{4})=1 | N(\frac{1}{2})=2)$



$$= \frac{P(N(\frac{1}{4})=1, N(\frac{1}{2})=2)}{P(N(\frac{1}{2})=2)}$$

↑ separables ↓ hipó condicional

$$= \frac{P(N(\frac{1}{4})=1) P(N(\frac{1}{4}, \frac{1}{2}))}{P(N(\frac{1}{2})=2)}$$

$$N(\frac{1}{4}) \sim \text{Poi}(\frac{1}{2})$$

$$N(\frac{1}{4}, \frac{1}{2}) \sim \text{Poi}(\frac{1}{2})$$

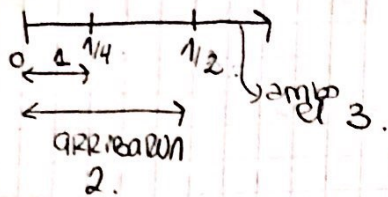
$$N(\frac{1}{2}) \sim \text{Poi}(1)$$

$$= \frac{(\frac{1}{2})^1 e^{-1/2}}{1!} \cdot \frac{(\frac{1}{2})^1 e^{-1/2}}{1!} = \frac{1^2 e^{-1}}{2!}$$

$$P(N(\frac{1}{4})=1 | S_3=\frac{1}{2}) = \frac{1}{2}$$

(2)

$$g) P(S_3 > 1/2 \mid N(1/4) = 1) = \frac{P(S_3 > 1/2, N(1/4) = 1)}{P(N(1/4) = 1)}$$



$$= \frac{P(N(0, 1/4) = 1, N(1/4, 1/2) \leq 1)}{P(N(0, 1/4) = 1)}$$

$$= \frac{P(N(0, 1/4) = 1) P(N(1/4, 1/2) \leq 1)}{P(N(0, 1/4) = 1)} \rightarrow \text{por ser indep.}$$

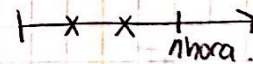
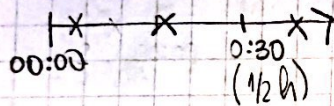
$$= \sum_{i=0}^1 \frac{(2 \cdot \frac{1}{4})^i}{i!} e^{-2 \cdot \frac{1}{4}} = e^{-\frac{1}{2}} + \frac{1}{2} e^{-\frac{1}{2}} = 0.9098$$

7.2

$N(t)$  = nº accidentes en intervalo

$$N \sim \text{Poi}(\lambda T) = \text{Poi}(2T) \quad \lambda = 2 \text{ por hora.}$$

a)



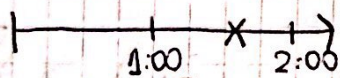
$$\mu = 2 \cdot \frac{1}{2} = 1 \quad m = 1$$

g1 en esa 1/2 hora solo pasan 2 accidentes  $P(S_3 > 1/2) =$  es lo mismo, usar gamma

$$P(N(0, 1/2) < 3) = \sum_{i=0}^2 \frac{\mu^i e^{-\mu}}{i!} = \frac{1^0 e^{-1}}{0!} + \frac{1^1 e^{-1}}{1!} + \frac{1^2 e^{-1}}{2!} = \frac{5}{2} e^{-1}$$

$$P(N(0, 1/2) < 3) = e^{-1} \cdot (1 + 1 + \frac{1}{2}) = \frac{5}{2} e^{-1}$$

b)



$$P(N(1, 2) = 1) = \frac{2^1 e^{-2}}{1!} = 2e^{-2}$$

NOTA

3

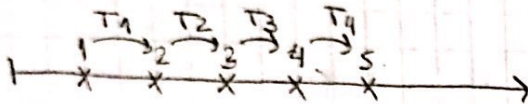
$\lambda = 10$  por hora

$N(t)$ : cantidad de partículas emitidas en ese tiempo

$N \sim \text{Poi}(\lambda t)$

$T$ : tiempo entre partículas consecutivas.

$T \sim \text{exp}(\lambda)$



Para que sean registradas, tiene que pasar  $\oplus$  de 2 minutos entre c/ partícula

60 min  $\rightarrow$  1 hora  
2 min  $\rightarrow$   $\frac{1}{30}$  hora.

$$P(T_1 > \frac{1}{30}, T_2 > \frac{1}{30}, T_3 > \frac{1}{30}, T_4 > \frac{1}{30}) =$$

como son todas expo e ind.

$$= P(T_1 > \frac{1}{30}) P(T_2 > \frac{1}{30}) P(T_3 > \frac{1}{30}) P(T_4 > \frac{1}{30})$$

$$= e^{-10 \cdot \frac{1}{30}} \cdot e^{-10/30} \cdot e^{-10/30} \cdot e^{-10/30}$$

$$= 0,2636 = P(\text{las primeras 5 sean detectadas})$$

PROB exp.

$$P(T_1 \leq \tau) = 1 - e^{-\lambda \tau}$$

$$P(T_1 > \tau) = e^{-\lambda \tau}$$



$$P(T_1 > \tau) = P(N(\tau) = 0) = \frac{(\lambda \tau)^0}{0!} e^{-\lambda \tau} = e^{-\lambda \tau}$$

4

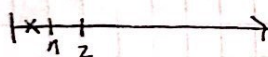
$\lambda = 2$  x seg

$\frac{\pi c}{\Delta \text{seg}}$   $\frac{\pi c}{\Delta \text{seg}}$

$\rightarrow$  solo se registra un roc.

$\mu = 2 \cdot \pi$

$R = \text{registram.}$



$$P(N(0,1) = 1, N(1,2) = 0)$$

$$P(R(1,2) = 0 \mid R(0,1) = 1) =$$

N. Señal emitida

son independientes

$$P(R(1,2) = 0, R(0,1) = 1) = 0,2077$$

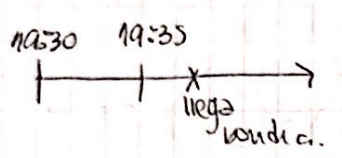
$$P(N(0,1) = 2, N(1,2) = 1) + P(N(0,1) = 2, N(1,2) = 0) + P(N(0,1) = 3, N(1,2) = 0)$$

$$P(N(0,1) = 2) + P(N(0,1) = 3)$$

$N(1,2) \sim \text{Poi}(2)$   
 $N(0,1) \sim \text{Poi}(2)$

4.5  $\lambda = 12 \times \text{hora}$

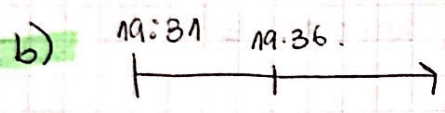
a) Andrés llega 19:30  $\rightarrow$  esperar + de 5 min.  
 5 min  $\rightarrow \frac{1}{12}$  hora.  
 60 min  $\rightarrow 1$  hora.



$T$ : tiempo hasta q' pasa el boudier

$$P(T > 5 \text{ min}) = P\left(T > \frac{1}{12} \text{ horas}\right) = e^{-12 \cdot \frac{1}{12}} = e^{-1}$$

$T \sim \text{exp}(12)$



$$P(T > \frac{1}{12}) = e^{-1}$$

c) Manas 19:30  $X = \{2, 3, 4, 5\}$  equiprob.

$$P(T > \frac{1}{12}) = e^{-1}$$

d) 19:00 + U  $U \sim \text{unif}(0,60)$

$$P(T > \frac{1}{12}) = e^{-1}$$

no importa horario que llegen, la probab. es por  $\oplus$  de 5 min es la misma.  
 $\hookrightarrow$  quiere decir que la probab. de que me vaya a perder la bolsa es la misma.

4.6  $\frac{1}{\lambda} = 3 \text{ kilos} \rightarrow \lambda = \frac{1}{3}$

$X$ : peso de bolsa pida.

a)  $X \sim \text{exp}(\frac{1}{3})$

$$S_m > 7$$

Yo se que supera los 5, y quiero que supere los 7.  
 si mi peso final  $Y$ : peso final

$$S_m = T_1 + T_2 + \dots$$

$$5 + \cancel{Y} > 7$$

$$P(Y > 2) = e^{-\frac{1}{3} \cdot 2} = e^{-\frac{2}{3}}$$

$$Y > 2$$

4) Si el evento O punto del ...

b)  $X_i \sim \text{Exp}(1/3)$

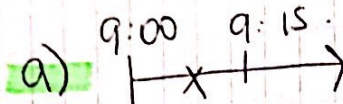
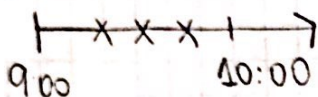
$E[X_i] = 3$

$X_i$  = cada bolsa  $X_i$  = última bolsa.

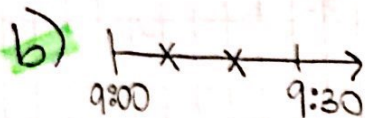
↓  
Todos los bolsos tienen la misma media.

15 min = 1/4 hora

f.f)  $\lambda = 4 \times \text{hora}$



Nº de llamadas q' llegan.



$P(N(0, 1/4) \geq 1 \mid N(0, 1) = 3) =$

$P(N(0, 1/2) \geq 2 \mid N(0, 1) = 3) = Y$  POR PROPIEDAD  $\rightarrow$  ③

$Y \sim \text{Bin}(3, \frac{1}{2})$

$X = N(0, 1/4) \mid N(0, 1) = 3 \sim \text{Bin}(3, \frac{1/4}{1})$

$P(Y \geq 2) = 1 - P(Y < 2) =$

$P(X \geq 1) = 1 - P(X < 1) =$

$1 - P(Y=0) - P(Y=1) =$

$1 - P(X=0) = 1 - \binom{3}{0} \left(\frac{1}{4}\right)^0 \left(1 - \frac{1}{4}\right)^3$

$1 - \binom{3}{0} \left(\frac{1}{2}\right)^0 \left(1 - \frac{1}{2}\right)^3 - \binom{3}{1} \left(\frac{1}{2}\right)^1 \left(1 - \frac{1}{2}\right)^2$

$1 - \frac{27}{64} = \boxed{0,58}$

$1 - \frac{1}{8} - \frac{3}{8} = \boxed{\frac{1}{2}}$

$\mu \exp(s)$

f.8)  $\lambda = 50 \times \text{minuto}$

$T$ : cantidad total de tiempo que esperan los pasajeros

$S_i$  = instante de llegada de cada pasajero  $E[T]$  y  $\text{Var}(T)$

$T_i$  = cantidad de tiempo q' espera el pasajero

$T = \sum_{i=1}^{N(0,15)} T_i$

$N(0, 15) = n \rightarrow$  cantidad de autobuses en 15 minutos

al condicionar  $N$  de  $(0, 15)$  se que

$$\boxed{T_i = 15 - S_i}$$

$\downarrow$  tiempo de espera       $\rightarrow$  tiempo llegado

$$T = \sum_{i=1}^{N(0,15)} (15 - S_i)$$

$$S_i \quad T_i = 15 - S_i \mid N(0,15) = m \sim U(0,15)$$

$$E[T] = E[E[T \mid N(0,15)]] =$$

$\downarrow$  PROP GUAR.S.

calculo aparte

$$E[T \mid N(0,15) = m] = E\left[\sum_{i=1}^m (15 - S_i) \mid N(0,15) = m\right]$$

$$= \sum_{i=1}^m \left(15 - E[S_i \mid N(0,15) = m]\right)$$

$U(0,15)$

$$= \sum_{i=1}^m \left(15 - \frac{15}{2}\right) = \frac{15}{2} m$$

$\frac{(a+b)}{2}$       valor que toma  $m$  veces.

$$E[T] = E\left[\frac{15}{2} m\right] = \frac{15}{2} E[m] =$$

$\text{según } N(0,15) = m$

$$= \frac{15}{2} E[N(0,15)] = \frac{15}{2} \mu = \frac{15}{2} \lambda \cdot 15$$

$$\downarrow$$

$$E[N] = \mu = \lambda \tau = \lambda \cdot 15$$

$$\boxed{E[T] = \frac{15}{2} \cdot 50 \cdot 15 = 5625}$$

$$\text{Var}(T) = \text{Var}\left(\underbrace{E[T \mid N(0,15)]}_{\text{I}}\right) + E\left(\underbrace{\text{Var}(T \mid N(0,15))}_{\text{II}}\right)$$

$\uparrow$  PITAGORAS.

$$\text{I} \quad \text{Var}\left(E[T \mid N(0,15)]\right) = \text{Var}\left(\frac{15}{2} m\right) = \left(\frac{15}{2}\right)^2 \cdot \text{Var}(m) \Rightarrow$$

$$\left(\frac{15}{2}\right)^2 \text{Var}(N) = \left(\frac{15}{2}\right)^2 \cdot \lambda \cdot 15 = \frac{84375}{2}$$

$$(ii) \text{Var}(T | N(0,15) = m) = \text{Var}\left(\sum_{i=1}^m (15 - S_i) | N(0,15) = m\right)$$

= como  $15 - S_i | N(0,15) = m$  son ind. y no aparecen covarianzas.

$$\sum_{i=1}^m \text{Var}(15 - S_i | N(0,15) = m) \rightarrow \text{solo } 15 \text{ afuera}$$

$$\sum_{i=1}^m \text{Var}(S_i | N(0,15) = m) = \sum_{i=1}^m \frac{15^2}{12} = \frac{75}{4} m$$

es uniforme.

$$\frac{(b-a)^2}{12}$$

$$E\left[\frac{75}{4} m\right] = \frac{75}{4} E[N(0,15)] = \frac{75}{4} \cdot \lambda \cdot 15$$

$$\text{Var}(T) = \frac{84375}{2} + \frac{28125}{2} = \boxed{56250}$$

7.9

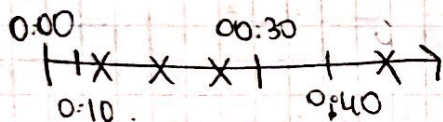
$\lambda = 6 \times \text{hora}$

$\Delta \rightarrow 60 \text{ min}$

$\frac{1}{6} \leftarrow \Delta 0 \text{ um}$

$\frac{2}{3} \leftarrow 40$

$\mu = \lambda T$



$$P([N(0,1/6) = 0 \cap N(0,2/3) = 3] | N(0,1/2) = 3) =$$

↓ de donde es, es ↓

$$P(N(0,1/6) = 0, N(1/6, 2/3) = 3) \stackrel{\downarrow \text{ por ser ind. dep.}}{=} \frac{P(N(0,1/2) = 3)}{P(N(0,1/6) = 0) P(N(1/6, 2/3) = 3)}$$

$$P(N(0,1/2) = 3)$$

$$\mu = 1 \leftarrow \frac{P(N(0,1/6) = 0) P(N(1/6, 2/3) = 3)}{P(N(0,1/2) = 3)} =$$

$$P(N(0,1/2) = 3) \rightarrow \mu = 6 \cdot \frac{1}{2} = 3$$

$$\left( \frac{1^0 e^{-1}}{0!} \cdot \frac{3^3 e^{-3}}{3!} \right) / \frac{3^3 e^{-3}}{3!} = 0.37$$

PROP 4 y 5.

↑  
7.10

↓ Fallas.

$$\lambda = 1 \text{ cada } 20 \text{ m} = \frac{1}{20} \text{ m}$$

PPP( $\lambda$ )

$$P(D) = 0.75$$

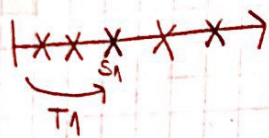
D: se detecta la falla

↓ se corta el alambre.

a) Long de rollos de alambre:  $X$  = hasta que se detecta la 1ra

$T_D \sim PP(\lambda_D = p\lambda) \rightarrow$  proceso del éxito  
↳ hallar falla

$T_1$ : la 1ra detectada



detectada



solo le doy  
importancia a las  
detectadas en el  
proceso

$$X = T_1 \sim \exp\left(\frac{p\lambda}{\lambda_1}\right)$$

$$E[X] = \frac{1}{p\lambda} = \frac{1}{(0.75) \frac{1}{20}} = \frac{80}{3}$$

$$\text{Var}(X) = \frac{1}{(p\lambda)^2} = \frac{1}{(0.75)^2 \frac{1}{20^2}} = \frac{6400}{9}$$

b) media de fallas en los rollos  $\rightarrow$  no detectados

las no detectadas  
sean de la ley.

$T_D \sim PP(\lambda_D = (1-p)\lambda) \rightarrow$  proceso de fracaso  
↳ no detectados.

$Y$ : no detectados

$$Y|X=x \sim \text{Poi}(\mu = \lambda_D \cdot x)$$

$$P(D) = 0.25$$

$$E[Y|X=x] = \frac{\mu}{(1-p)\lambda \cdot x}$$

$$E[Y] = E[E[Y|X=x]] = E[(1-p)\lambda x] = (1-p)\lambda E[X] =$$

↓  
las no  
detectadas.

$$= \frac{(1-0.75)}{20} \cdot \frac{80}{3} = \frac{1}{80} \cdot \frac{80}{3} = \boxed{\frac{1}{3}}$$

f.11

$$\lambda = \frac{1}{20} \text{ m}$$

$$P_{\text{falla der.}} = 0.75$$

CORTA  $\rightarrow$  1ra detectada antes de los 20 o a los 20 m.

a) media  $X$ : tiempo hollos. [m]

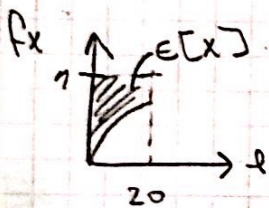
$$X = \begin{cases} T_1^D, & T_1^D < 20 \\ 20, & T_1^D \geq 20 \end{cases}$$

$T_1^D$ : 1ra falla hollos

$$T_1^D \sim \exp(\lambda)$$

$$\begin{aligned} E[X] &= E[X | T_1^D < 20] P(T_1^D < 20) + E[X | T_1^D \geq 20] P(T_1^D \geq 20) \\ &\downarrow \text{uso la formula de esperanza total y repito los 2 casos.} \\ &= E[T_1^D | T_1^D < 20] (1 - e^{-\lambda \cdot 20}) + E[20 | T_1^D \geq 20] \cdot e^{-\lambda \cdot 20} = \end{aligned}$$

OTRA FORMA.



$\rightarrow$  como es def positiva (no toma valores neg) y lo puedo relacionar como el área sobre la  $f$  densidad

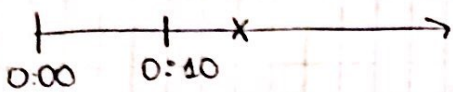
$$\begin{aligned} E[X] &= \int_0^{20} 1 - (1 - e^{-\lambda x}) dx = \\ &= \int_0^{20} (1 - e^{-\frac{1}{20}x}) dx = 14.07 \end{aligned}$$

b) Hallar la media de los follos en hollos  $\rightarrow$  igual que antes.

$$E[N] = (1 - P)(\lambda) E[X] = \frac{1}{20} (14.07) = \boxed{0.7035}$$

5

7.12 Lucas  $\rightarrow$  PPP( $\lambda_1$ )  $\lambda_1 = 3$  minutos. } independientes.  
 Monk  $\rightarrow$  PPP( $\lambda_2$ )  $\lambda_2 = 5$  minutos.

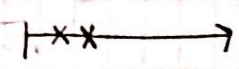
a)   $\pi_1 + \pi_2 \sim \text{PPP}(\lambda_1 + \lambda_2)$

superposición

$$T_k \sim \text{exp}(\lambda_1 + \lambda_2)$$

$$P(T_1 > 10) = e^{-(\lambda_1 + \lambda_2) \cdot 10} = e^{-80}$$

b)  $U = \min \{ T_1^{(1)}, T_1^{(2)} \}$

 solo voy a considerar el  
 por ambos de c/ proceso.  
 si yo quiero 10 sea de Lucas.  
 $U = T_1^{(1)}$   $J_1 \begin{cases} 1 = \text{Lucas} \\ 0 = \text{Monk} \end{cases}$

$$P(T_1^{(1)}) = P(J_1 = 1) = \frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{3}{8}$$

$\downarrow$   
 $P(T_L < T_M)$

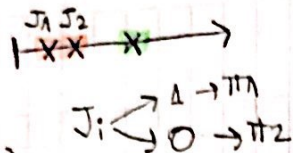
c) uno todo como  $U$  y  $J$  son ind x el 4.14

$$P(T_1^{(1)} > 10 \cap J=1) = P(T_1^{(1)} > 10) P(J=1)$$

$$U \sim \text{exp}(\lambda_1 + \lambda_2)$$

$$= e^{-(\lambda_1 + \lambda_2) \cdot 10} \cdot \frac{\lambda_1}{\lambda_1 + \lambda_2} = \boxed{e^{-80} \cdot \frac{3}{8}}$$

4.13  $\pi_1 \sim PP(2)$   
 $\pi_2 \sim PP(2)$



uso el 5 de 4.14

a)  $P(J_1=1 \cap J_2=1) =$   
 como el evento es ind

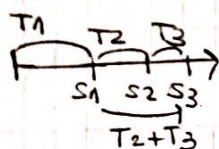
$$P(J_1=1) P(J_2=1) = \frac{\lambda_1}{\lambda_1 + \lambda_2} \cdot \frac{\lambda_2}{\lambda_1 + \lambda_2} = \frac{2}{4} \cdot \frac{2}{4} = \frac{4}{16}$$

$$(J_1=1) = (T_1^{pi_1} < T_1^{pi_2})$$

$$(J_2=1) = (T_2^{pi_2} < T_2^{pi_1})$$

b)  $\lambda = 2 \quad E[exp] = \frac{1}{\lambda} = \frac{1}{2}$

tiempo desde 1ra señal



$$T_1 > T_2 + T_3$$

tiempo hasta 1ra señal

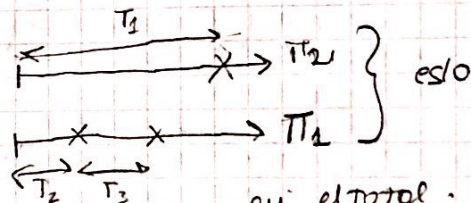
$$T_1 \sim E(1/2)$$

$$T_2 \sim E(1/2)$$

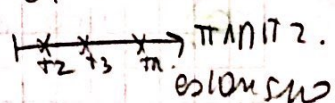
$$T_3 \sim E(1/2)$$

si yo quiero que el tiempo hasta la primera sea mayor que hasta  $T_2$  y  $T_3$ .

$$P(T_1 > T_2 + T_3) = 1/4$$



en el total.

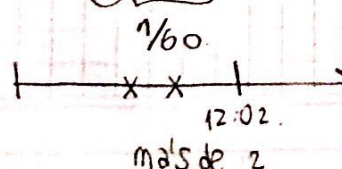
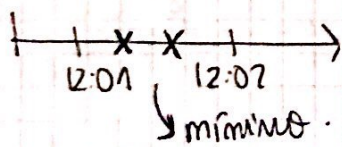
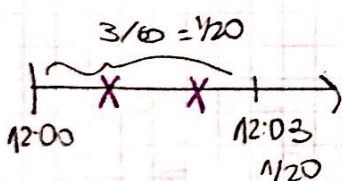


4.14

$$\lambda = 60 \text{ x hora} \rightarrow \text{patrujas}$$

$$\lambda = 40 \text{ x hora} \rightarrow \text{chisperos}$$

ind



si superpongo los pp.

$$PP(\lambda = 100 \text{ x hora})$$

$$N = N_1 + N_2$$

$N$ : cont de ambos de patrullas.

$N$ : cont de ambos de chisperos.

$$P(N(1/60, 1/30) \geq 2 | N_2(1/20) = 2) =$$

defino

$$N(1/60, 1/30) | N_2(1/20) = 2 \sim \text{Bin}(2, 1/3)$$

$$P(T \geq t) = \sum_{n=0}^{k-1} \frac{e^{-\lambda t} (\lambda t)^n}{n!}$$

$$\rightarrow \text{superpongo los pp.} \quad \frac{1/60 + 1/60}{1/20}$$

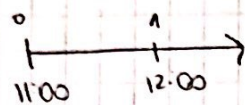
$$Sp = 1 - F(T) \quad P(T \geq 2) = \sum_{n=0}^{\infty} \frac{e^{-\frac{1}{2} \cdot 2} (\frac{1}{2} \cdot 2)^n}{n!} \approx 0,8557 \quad \leftarrow 7/11$$

7.15.

PPP (4) x hora.

$X$ : tempo de trabalho.

$$X \sim N(\frac{5}{12}, 1/2) \quad E(X) \text{ y } Var(X) \text{ en me 11 1/2.}$$



$$T_T = \text{tempo total} = X = \sum_{i=1}^{N(0,1)=m} T_i$$

$$E[T] = E[E[T | N(0,1)=m]]$$

$$= E[E[\sum_{i=1}^m T_i]] = E[E[mT]]$$

$$m E[T] = m \cdot \frac{5}{60}$$

$$E[m \frac{5}{60}] = \frac{5}{60} E[N(0,1)] = \boxed{\frac{5}{60} \cdot 4 \text{ h.}}$$

μ poisson.

$$V(T_T) = Var(E(T_T | N(0,1))) + E(Var(T_T | N(0,1))) \rightarrow \text{igual 7.8.}$$

7.16.

$\lambda = 3.000$  x semana.

$$\pi \sim PPP(\lambda)$$

↓ familias que migran

$$\begin{matrix} 3 & p=0,15 \\ 4 & p=0,4 \\ 5 & p=0,1 \end{matrix}$$

Nº integrantes

Nº cantidad q' migran en 4 semanas.

voy a dividir el PP gral en 3 dependiendo la familia.

$$\pi_1 \sim PPP(\lambda_1 = 0,15 \cdot \lambda) \rightarrow 3 \text{ integrantes}$$

$$\pi_2 \sim PPP(\lambda_2 = 0,4 \cdot \lambda) \rightarrow 4 \quad \text{"}$$

$$\pi_3 \sim PPP(\lambda_3 = 0,1 \cdot \lambda) \rightarrow 5 \quad \text{"}$$

NOTA

$$N = 3 N_1 (0,4) + 4 N_2 (0,4) + 5 N_3 (0,4)$$

↓  
cont g' migrau de semana 10 a 4  
del  $\pi_1$

$$E[N] = 3 E[N_1 (0,4)] + 4 E[N_2 (0,4)] + 5 E[N_3 (0,4)]$$

$$N_i \sim \text{Poi} (\mu = \lambda_i \cdot 4)$$

$$E[N] = 3 \cdot \underset{0,157}{\lambda_1 \cdot 4} + 4 \underset{0,47}{\lambda_2 \cdot 4} + 5 \cdot \underset{0,117}{\lambda_3 \cdot 4}$$

$$\boxed{E[N] = 43200}$$

como el proceso es indep  $\rightarrow \text{cov} = 0 \rightarrow$  la var es la suma de las var

$$\text{var}(N) = 3^2 \text{var}(N_1 (0,4)) + 4^2 \text{var}(N_2 (0,4)) + 5^2 \text{var}(N_3 (0,4))$$

$$\text{var}(N) = 9 \cdot \lambda_1 4 + 16 \lambda_2 4 + 25 \lambda_3 4$$

$$\boxed{\text{var}(N) = 160800}$$

Ex 7.17

PPP (10 x minutos)  $\pi$ : vehiculos amikar.

1) 10% = 0,1 autos

$X$ : peso de carga.

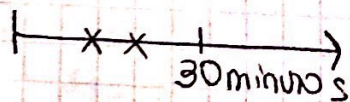
2) 10% = 0,1 motos.

$$E[X_1] = 400$$

3) 20% = 0,2 camiones.

$$E[X_2] = 120$$

$$E[X_3] = 1300$$



$$P(N_1 (0,1/2) \geq 2) = 1 - P(N(0,1/2) < 2)$$

$$1 - P(N(0,1/2) = 0) - P(N(0,1/2) = 1) =$$

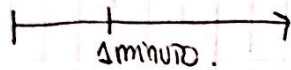
$$\mu = \lambda \cdot p_2 \cdot \tau$$

8

$$= 1 - \frac{(10 \cdot 0,7 \cdot \frac{1}{2})^0 e^{-10 \cdot 0,7 \cdot \frac{1}{2}}}{0!} - \frac{(\frac{7}{2})^1 e^{-7/2}}{1!}$$

$$= \boxed{0,86}$$

b)



$$\mu_1 = \lambda p_1 \frac{1}{60} = \frac{7}{60} \quad \mu_2 = \lambda p_2 \frac{1}{60} = \frac{1}{60}$$

$$\mu_3 = \lambda p_3 \frac{1}{60} = \frac{1}{30}$$

$$P(N_1(0, 1/60) = 7, N_2(0, 1/60) = 1, N_3(0, 1/60) = 2) =$$

como  
são indep.

$$\frac{(\frac{7}{60})^7 e^{-\frac{7}{60}}}{7!} \cdot \frac{(\frac{1}{60})^1 e^{-\frac{1}{60}}}{1!} \cdot \frac{(\frac{1}{30})^2 e^{-\frac{1}{30}}}{2!}$$

$$P(\dots) = 4,6 \times 10^{-16}$$

c)

formula de E total

$$E(x) = E x_1 \cdot p_{x1} + E x_2 \cdot p_{x2} + E x_3 \cdot p_{x3}$$

$$= 400 \cdot 0,7 + 0,1 \cdot 120 + 1300 \cdot 0,2 = 552$$

d)

T: carga total em 1 hora.

$$E(T) = 400 \text{ kg } E[N_1(0,60)] + 120 E[N_2(0,60)] + E[N_3(0,60)] 1300 \text{ kg} =$$

$$= 400 \cdot 420 + 120 \cdot 60 + 1300 \cdot 120$$

$$\boxed{E(T) = 331200 \text{ kg}}$$

NOTA

7.18)

A 2 x minuto B 1 x minuto.  $\rightarrow$  md.

$$P(\text{par}) = P(J_1=1) P(J_2=2) = \frac{\lambda_A}{\lambda_A + \lambda_B} \cdot \frac{\lambda_B}{\lambda_B + \lambda_A}$$
$$= \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{9}$$

T: tiempo hasta que pase uno.